

Lecture 2. Statistical Schools of Thought

COMP90051 Statistical Machine Learning

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This lecture

How do learning algorithms come about?

- Frequentist statistics
- Statistical decision theory
- Bayesian statistics

Types of probabilistic models

- Parametric vs. Non-parametric
- Generative vs. Discriminative

Frequentist Statistics

Wherein unknown model parameters are treated as having fixed but unknown values.

Frequentist statistics

- Abstract problem

- * Given: $X_1, X_2, \dots, X_n$ drawn i.i.d. from some distribution
- * Want to: identify unknown distribution, or a property of it

Independent and
identically distributed

- Parametric approach (“**parameter estimation**”)

- * Class of **models** $\{p_{\theta}(x): \theta \in \Theta\}$ indexed by **parameters** in Θ (could be a real number, or vector, or)
- * **Point estimate** $\hat{\theta}(X_1, \dots, X_n)$ a function (or **statistic**) of data

- Examples

- * Given n coin flips, determine probability of landing heads
- * Learning a classifier

Hat means estimate
or estimator

Estimator Bias

Frequentists seek good behaviour, in ideal conditions

- **Bias:** $B_{\theta}(\hat{\theta}) = E_{\theta}[\hat{\theta}(X_1, \dots, X_n)] - \theta$

Example: for $i=1\dots 40$

- $X_{i,1}, \dots, X_{i,20} \sim p_{\theta} = \text{Normal}(\theta = 0, \sigma^2 = 1)$

Subscript θ means data
really comes from p_{θ}

Estimator Bias

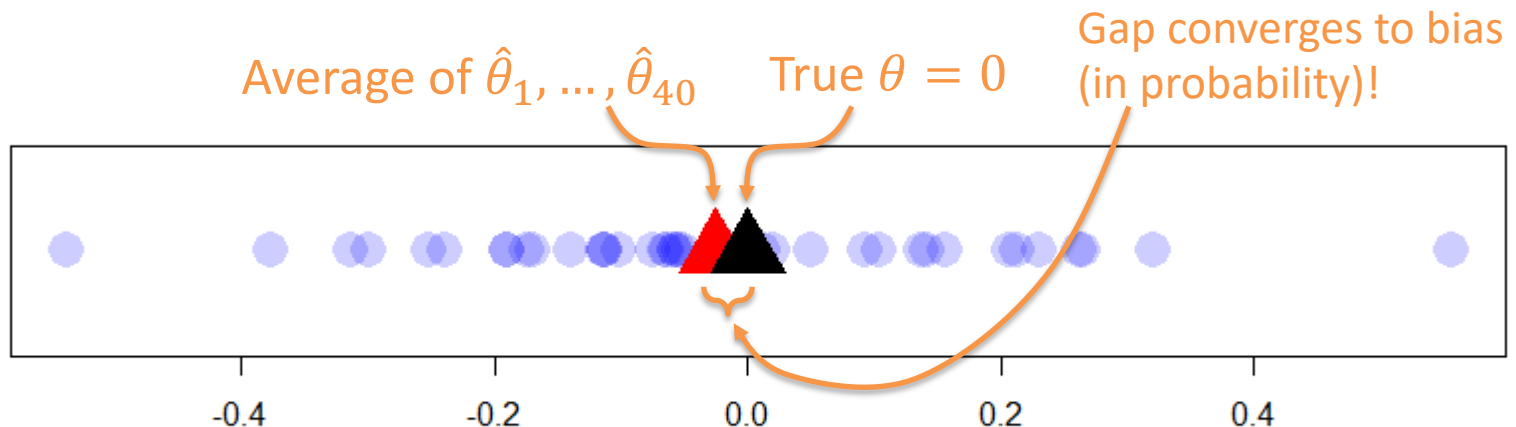
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Subscript θ means data really comes from p_{θ}

Example: for $i=1\dots 40$

- $X_{i,1}, \dots, X_{i,20} \sim p_{\theta} = \text{Normal}(\theta = 0, \sigma^2 = 1)$
- $\hat{\theta}_i = \frac{1}{20} \sum_{j=1}^{20} X_{i,j}$ the sample mean, plot as ●



Estimator Variance

Frequentists seek good behaviour, in ideal conditions

- **Variance:** $\text{Var}_\theta(\hat{\theta}) = \text{E}_\theta[(\hat{\theta} - \text{E}_\theta[\hat{\theta}])^2]$

$\hat{\theta}$ still function of data

Estimator Variance

Frequentists seek good behaviour, in ideal conditions

- **Variance:** $\text{Var}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \mathbb{E}_\theta[\hat{\theta}])^2]$

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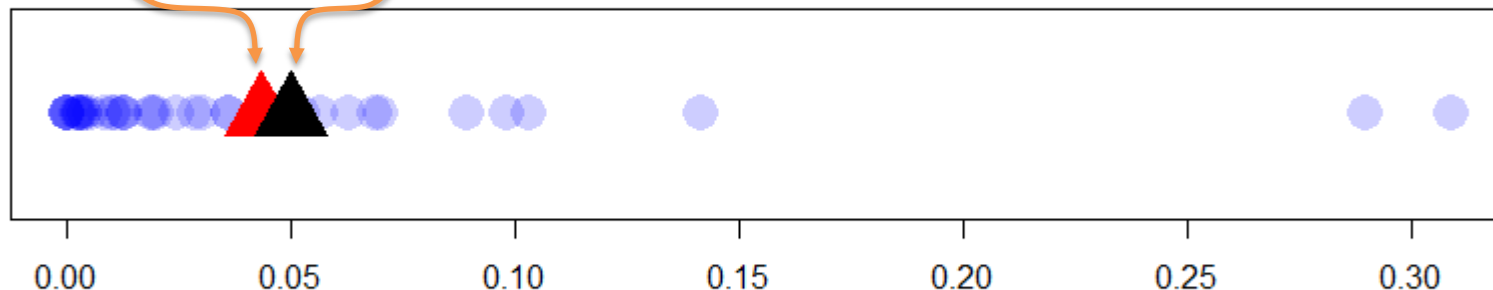
Example cont.

- Plot each $(\hat{\theta}_i - \mathbb{E}_\theta[\hat{\theta}_i])^2 = \hat{\theta}_i^2$ as ●

Average of $\hat{\theta}_1^2, \dots, \hat{\theta}_{40}^2$

True $\text{Var}_\theta(\hat{\theta}) = \frac{\sigma^2}{20} = 0.05$

Once again, average converges to true (in probability)!



Asymptotically Well Behaved

For our example estimator (sample mean), we could calculate its exact bias (zero) and variance (σ^2). Usually can't guarantee low bias/variance exactly 😞

Asymptotic properties often hold! 😊

- **Consistency:** $\hat{\theta}(X_1, \dots, X_n) \rightarrow \theta$ in probability
- **Asymptotic efficiency:** $\text{Var}_{\theta} \left(\hat{\theta}(X_1, \dots, X_n) \right)$ converges to the smallest possible variance of any estimator of θ

Bias closer and closer to zero

Variance closer & closer to optimal

Maximum-Likelihood Estimation

- A **general principle** for designing estimators
- Involves **optimisation**
- $\hat{\theta}(x_1, \dots, x_n) \in \operatorname{argmax}_{\theta \in \Theta} \prod_{i=1}^n p_{\theta}(x_i)$
- *“The best estimate is one under which observed data is most likely”*



Fischer

Example I: Bernoulli

- Know data comes from Bernoulli distribution with unknown parameter (e.g., biased coin); find mean
- MLE for mean

$$* p_{\theta}(x) = \begin{cases} \theta, & \text{if } x = 1 \\ 1 - \theta, & \text{if } x = 0 \end{cases} = \theta^x (1 - \theta)^{1-x}$$

(note: $p_{\theta}(x) = 0$ for all other x)

$$* \text{ Maximising likelihood yields } \hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$$

Example I: Bernoulli

1. Write as joint distribution over all X

$$\prod_{i=1}^n p_{\theta}(x_i) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}$$

2. Take logarithms

$$\begin{aligned} \log \prod_{i=1}^n p_{\theta}(x_i) &= \sum_{i=1}^n x_i \log \theta + (1 - x_i) \log(1 - \theta) \\ &= L(\theta) \end{aligned}$$

3. Find derivative. Set $\sum_{i=1}^n x_i = \bar{X}$

$$\frac{d}{d\theta} L(\theta) = \frac{\bar{X}}{\theta} - \frac{n - \bar{X}}{1 - \theta}$$

4. Set equal to zero and solve for $\hat{\theta}$:

$$\hat{\theta} = \frac{\bar{X}}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example II: Normal

- Know data comes from Normal distribution with variance 1 but unknown mean; find mean

- MLE for mean

- * $p_{\theta}(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x - \theta)^2\right)$

- * Maximising likelihood yields $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$

- Exercise: derive MLE for *variance* σ^2 based on

$$p_{\theta}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) \text{ with } \theta = (\mu, \sigma^2)$$

MLE 'algorithm'

1. Given data $X_1, \dots, X_n$ define probability distribution, p_{θ} , assumed to have generated the data
2. Express likelihood of data, $\prod_{i=1}^n p_{\theta}(X_i)$
(usually its *logarithm*... why?)
3. Optimise to find *best* (most likely) parameters $\hat{\theta}$
 1. take partial derivatives of log likelihood wrt θ
 2. set to 0 and solve
(failing that, use gradient descent)

Mini Summary

- Frequentist school of thought
- Point estimates
- Quality: bias, variance, consistency, asymptotic efficiency
- Maximum-likelihood estimation (MLE)

Next: Statistical Decision Theory

Statistical Decision Theory

Branch within statistics, optimisation, economics, control, emphasising utility maximisation.

Decision theory



Wald

- Act to maximise utility - connected to economics and operations research
- **Decision rule** $\delta(\mathbf{x}) \in A$ an action space
 - * E.g. Point estimate $\hat{\theta}(x_1, \dots, x_n)$
 - * E.g. Out-of-sample prediction $\hat{Y}_{n+1} | X_1, Y_1, \dots, X_n, Y_n, X_{n+1}$
- **Loss function** l : economic cost, error metric
 - * E.g. square loss of estimate $l(\hat{\theta}, x) = (\hat{\theta} - x)^2$
 - * E.g. 0-1 loss of classifier predictions $l(\hat{y}, y) = 1[\hat{y} \neq y]$

Risk & Empirical Risk Minimisation (ERM)

- In decision theory, really care about *expected* loss
- **Risk** $R_\theta[\delta] = \mathbb{E}_{\mathbf{X} \sim \theta}[l(\delta, \mathbf{X})]$ or

$$R_\theta[\delta] = \mathbb{E}_{(\mathbf{X}, Y) \sim \theta}[l(\delta(\mathbf{X}), Y)]$$

* E.g. true test error, aka generalization error

- Want: Choose δ to minimise $R_\theta[\delta]$
- Can't directly! Why?
- **ERM**: Use training set $\mathbf{X}$ to approximate R_θ
 - * Minimise **empirical risk** $\hat{R}_\theta[\delta] = \frac{1}{n} \sum_{i=1}^n l(\delta, X_i)$ or
$$\hat{R}_\theta[\delta] = \frac{1}{n} \sum_{i=1}^n l(\delta(X_i), Y_i)$$

Decision theory vs. Bias-variance

We've already seen

- Bias: $B_{\theta}(\hat{\theta}) = E_{\theta}[\hat{\theta}(X_1, \dots, X_n)] - \theta$
- Variance: $\text{Var}_{\theta}(\hat{\theta}) = E_{\theta}[(\hat{\theta} - E_{\theta}[\hat{\theta}])^2]$

But are they equally important? How related?

- **Bias-variance decomposition** of square-loss risk

$$E_{\theta}[(\theta - \hat{\theta})^2] = [B(\hat{\theta})]^2 + \text{Var}_{\theta}(\hat{\theta})$$

Bayesian Statistics

Wherein unknown model parameters have associated distributions reflecting prior belief.

Bayesian statistics



Laplace

- Probabilities correspond to **beliefs**
- Parameters
 - * Modeled as r.v.'s having distributions
 - * Prior belief in θ encoded by **prior distribution** $P(\theta)$
 - Parameters are modeled like r.v.'s (even if not really random)
 - Thus: data likelihood $P_{\theta}(X)$ written as conditional $P(X|\theta)$
 - * Rather than point estimate $\hat{\theta}$, Bayesians update belief $P(\theta)$ with observed data to $P(\theta|X)$ the **posterior distribution**

Tools of probabilistic inference

- Bayesian probabilistic inference

- * Start with prior $P(\theta)$ and likelihood $P(X|\theta)$
- * Observe data $X = x$
- * Update prior to posterior $P(\theta|X = x)$



Bayes

- Primary tools to obtain the posterior

- * **Bayes Rule**: reverses order of conditioning

$$P(\theta|X = x) = \frac{P(X = x|\theta)P(\theta)}{P(X = x)}$$

- * **Marginalisation**: eliminates unwanted variables

$$P(X = x) = \sum_t P(X = x, \theta = t)$$

This quantity
is called the
evidence

These are
general tools of
probability and
not specific to
Bayesian
stats/ML

Example

- **Given:** The data comes from a **Normal distribution with variance 1 but unknown mean θ** .
- **Goal:** Find the **posterior over the mean** after seeing one data point where **$X=1$** . Assume a **Normal prior over θ with mean 0 and variance 1**.

$$P(\theta|X = 1) = \frac{P(X = 1|\theta)P(\theta)}{P(X = 1)}$$

$$\propto P(X = 1|\theta)P(\theta)$$

$$= \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1-\theta)^2}{2}\right) \right] \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\theta^2}{2}\right) \right]$$

$$\propto \exp\left(-\frac{(1-\theta)^2 + \theta^2}{2}\right) = \exp\left(-\frac{2\theta^2 - 2\theta + 1}{2}\right)$$

$$= \exp\left(-\frac{\theta^2 - \theta + \frac{1}{2}}{2 \times \frac{1}{2}}\right)$$

The Normal distribution:

$$\mathcal{N}(x; \mu, \sigma^2) \equiv \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Discard constants
wrt θ

To have θ^2 in numerator instead of $2\theta^2$:
 $\times \frac{1}{2}$ on top and bottom

Example

$$P(\theta|X=1) = \exp\left(-\frac{\theta^2 - \theta + \frac{1}{2}}{2 \times \frac{1}{2}}\right)$$

$$= \exp\left(-\frac{(\theta - \frac{1}{2})^2 + \frac{1}{4}}{2 \times \frac{1}{2}}\right)$$

$$= \exp\left(-\frac{(\theta - \frac{1}{2})^2}{2 \times \frac{1}{2}}\right) \cdot \exp\left(-\frac{\frac{1}{4}}{2 \times \frac{1}{2}}\right)$$

$$\propto \exp\left(-\frac{(\theta - \frac{1}{2})^2}{2 \times \frac{1}{2}}\right)$$

$$\propto \mathcal{N}(0.5, 0.5)$$

The square completion trick!

https://en.wikipedia.org/wiki/Completing_the_square

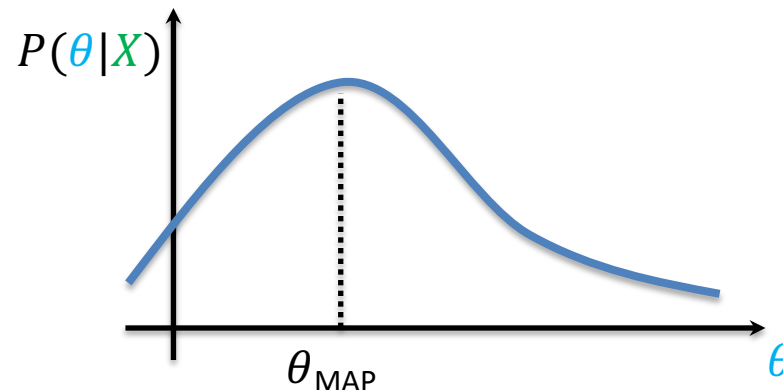
$$\begin{aligned}\theta^2 - \theta + \frac{1}{2} &\rightarrow (\theta - ?)^2 + ? \\ \theta^2 - \theta + \frac{1}{2} &= (\theta - \frac{1}{2})^2 + \frac{1}{4}\end{aligned}$$

The Normal distribution:

$$\mathcal{N}(x; \mu, \sigma^2) \equiv \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

How Bayesians make point estimates

- They don't, unless forced to do so!
 - * The posterior carries full information, why discard it?
- But, there are common approaches
 - * Posterior mean $E_{\theta|X}[\theta] = \int \theta P(\theta|X) d\theta$
 - * Posterior mode $\underset{\theta}{\operatorname{argmax}} P(\theta|X)$ (**max a posteriori** or MAP)
 - * There're Bayesian decision-theoretic interpretations of these



MLE in Bayesian context

- **MLE** formulation: find parameters that best fit data
$$\hat{\theta} \in \operatorname{argmax}_{\theta} P(X = x|\theta)$$
- Consider the **MAP** under a Bayesian formulation
$$\begin{aligned}\hat{\theta} &\in \operatorname{argmax}_{\theta} P(\theta|X = x) \\ &= \operatorname{argmax}_{\theta} \frac{P(X = x|\theta)P(\theta)}{P(X = x)} \\ &= \operatorname{argmax}_{\theta} P(X = x|\theta)P(\theta)\end{aligned}$$
- **Prior** $P(\theta)$ weights; MLE like *uniform* $P(\theta) \propto 1$

Frequentists vs Bayesians – Oh My!

- Two key schools of statistical thinking
 - * Decision theory complements both
- Past: controversy; animosity; almost a 'religious' choice
- Nowadays: deeply connected

I declare the Bayesian vs. Frequentist debate over for data scientists

👤 Rafael Irizarry 📅 2014/10/13

Are You a Bayesian or a Frequentist?

Michael I. Jordan
 Department of EECS
 Department of Statistics
 University of California, Berkeley

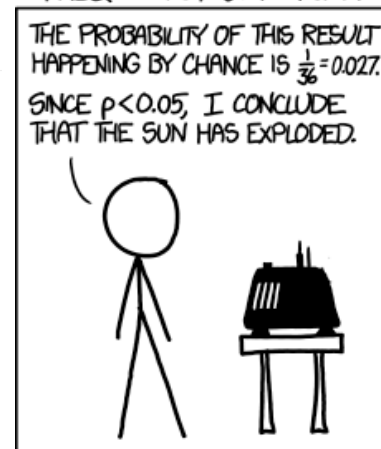
<http://www.cs.berkeley.edu/~jordan>

DID THE SUN JUST EXPLODE?
 (IT'S NIGHT, SO WE'RE NOT SURE.)

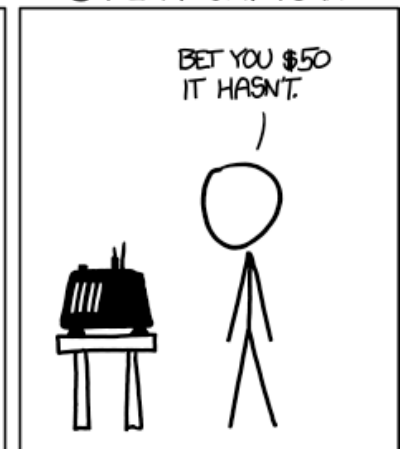


<https://xkcd.com/1132/> CC-NC2.5

FREQUENTIST STATISTICIAN:



BAYESIAN STATISTICIAN:



(Some) Categories of Probabilistic Models

Parametric vs non-parametric models

Parametric	Non-Parametric
Determined by fixed, finite number of parameters	Number of parameters grows with data, potentially infinite
Limited flexibility	More flexible
Efficient statistically and computationally	Less efficient

Examples to come! There are non/parametric models in both the frequentist and Bayesian schools.

Generative vs. discriminative models

- X 's are instances, Y 's are labels (**supervised setting!**)
 - * Given: i.i.d. data $(X_1, Y_1), \dots, (X_n, Y_n)$
 - * Find model that can predict Y of new X
- Generative approach
 - * Model full joint $P(X, Y)$
- Discriminative approach
 - * Model conditional $P(Y|X)$ only
- Both have pro's and con's

Examples to come! There are generative/discriminative models in both the frequentist and Bayesian schools.

Mini Summary

- Bayesian paradigm: It's all in the prior!
- Bayesian point estimate: MAP
- Parametric vs Non-parametric models
- Discriminative vs. Generative models

Next: Linear regression (unlike you've ever seen before)